Mid-Term Exam : Climate Finance

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Exam totals 75 points. Please write at most one line per point to answer a question.

1 Problem: 20 points

Recall that the Hill index of biodiversity of order q is given by:

$$D\{q\} = \left(\sum_{i=1}^{S} p_i^q \right)^{\frac{1}{1-q}} \tag{1}$$

Assume that there are two species and write $p \leq \frac{1}{2}$ for the proportion of the rarest species and $D^p\{q\}$ for Hill index of order q when the proportion of rare species is p.

(a) Show that the Hill index for $q = \frac{1}{2}$ increases with $p \leq \frac{1}{2}$ but is always $\leq$ than the species count. Give an intuitive explanation why it makes sense to use the diversity index expressed by the Hill index for $q = \frac{1}{2}$, even though species count does not vary.

(b) Assume that one species is light demanding and the other is shade tolerant. Consider an area of secondary tropical forest that started reforesting at $t = 0$. Which species would be rarer for small t?

(c) Suppose that if the area was virgin forest then the proportion of the two species would be the same. Draw two curves that describe the evolution you would expect for species count and for the Hill index of order $q = \frac{1}{2}$.

(d) What are the implication, if any, of your answer to item (c), to possible externalities generated by deforestation in a site on other sites.

2 Problem: 20 points

While earlier literature on the effect of temperature on economic outcomes used shocks to local temperature, Bilal and Kanzig show that shocks to global temperature are better predictors of economic outcomes than abnormal local temperature.

(a) Give one example of an event that affect global temperature and can cause large economic effects, while not necessarily affecting local temperature. Please explain.

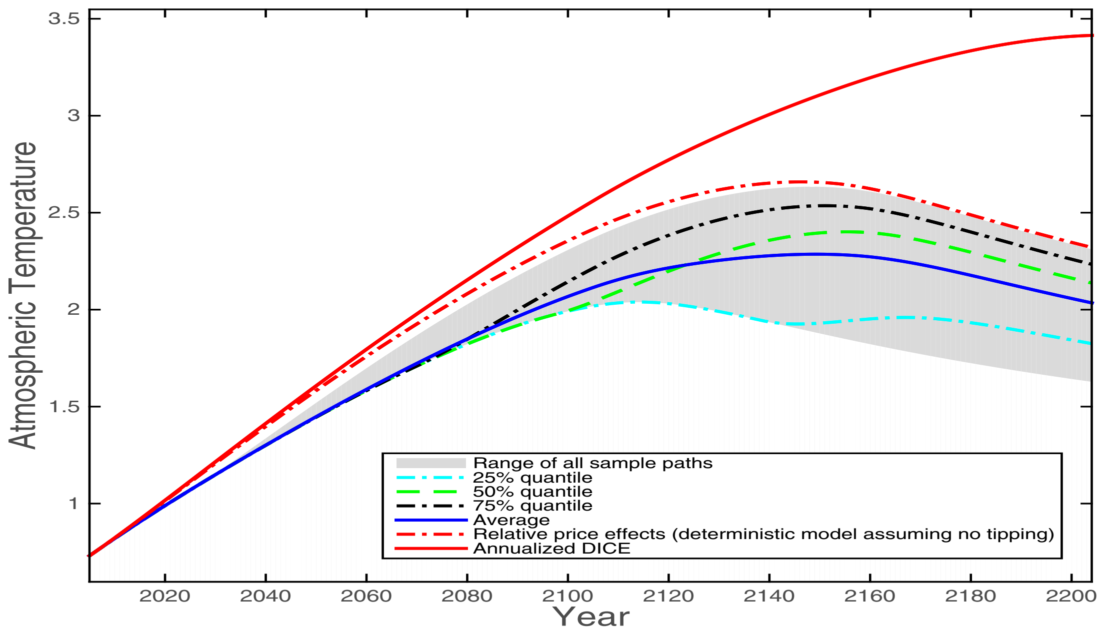
(b) Describe using equations or words how Bilal and Kanzig constructed a measure of global temperature shocks and how they inferred the effect of global temperature shocks on economic outcomes.

(c) How different are the predictions for the effect of temperature rise on high income countries when you use local vs. global temperature shocks.

(d) How would the predictions you discussed on item (c) influence negotiations on financing emissions reductions.

3 Problem: 15 points

The Figure below plots curves of the surface temperature implied by the optimal solution of different models: the red curve uses Dice and no tipping point. All other curves assume an elasticity of substitution between environmental and other consumption goods of .5. The red dashed curve assumes no tipping point, all other curves assume the possibility of a tipping point that discretely lowers the supply of the environmental good .



(a) In the figure above explain the relative positions of the solid red and dashed red curves.

(b) Explain the the relative positions of the solid blue and dashed red curves.

(c) Explain the abrupt change in the derivative of the cyan curve (25% percentile) around year 2120.

4 True, False or Uncertain - justify your answer in less than 5 lines (5 points each).

1. Every country should finance the reduction of its own carbon footprint.
2. Since as forests mature, total carbon capture diminishes, any agreement to induce reforestation by paying for carbon capture is not sustainable in the long run.
3. Proven reserves of oil go down as oil is consumed and thus the price of oil increases over time.

4. Satellite data can been effectively used to deter illegal deforestation in tropical forests.